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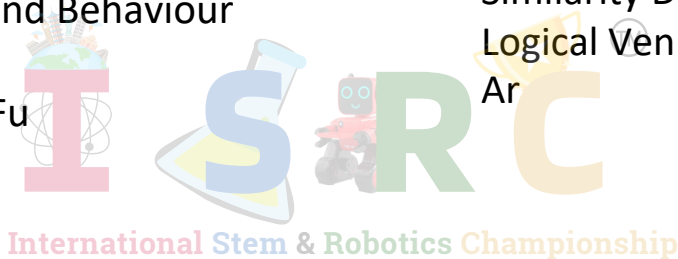
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1.English

1.1 Tense

Tense indicates the time of an action or state of being. Understanding tense forms helps convey the correct time frame in writing and speech.

Key Tenses:

1. **Present Tense:** Indicates actions happening now.
o Example: She **writes** a letter.
2. **Past Tense:** Indicates actions that have already occurred.
o Example: He **wrote** a letter.
3. **Future Tense:** Indicates actions that will happen.
o Example: They **will write** a letter.

Questions:

1. Convert the following sentence to past tense: "She walks to school."
2. Write a sentence in future tense.
3. Identify the tense in this sentence: "They are playing soccer."
4. Provide an example of a sentence in simple present tense.

1.2 Modal

Modals are auxiliary verbs that express necessity, possibility, permission, or ability. They modify the meaning of the main verb.

Common Modals:

- ☐ **Can:** Indicates ability (e.g., She **can** swim).
- ☐ **Could:** Indicates past ability or possibility (e.g., He **could** play the guitar).
- ☐ **May:** Indicates permission or possibility (e.g., You **may** leave early).
- ☐ **Must:** Indicates necessity (e.g., You **must** finish your homework).

Questions:

1. Write a sentence using "can" to express ability.
2. How is "must" different from "may"?
3. Provide an example of "should" used in a sentence.
4. Convert the following: "He is able to run fast" using a modal.

1.3 Subject-Verb Concord

Subject-verb concord (or agreement) means that the subject and verb in a sentence must agree in number and person.

Key Concepts:

- ☐ **Singular Subjects:** Take singular verbs (e.g., The cat **sleeps**).
- ☐ **Plural Subjects:** Take plural verbs (e.g., The cats **sleep**).
- ☐ **Indefinite Pronouns:** Some require singular verbs (e.g., Everyone **is** happy).

Questions:

1. Identify the error in the following sentence: "The team are winning the match."
2. Write a correct sentence using a plural subject and verb.
3. Explain the rule of concord with collective nouns.
4. Provide an example of a sentence using an indefinite pronoun.

1.4 Reported Speech

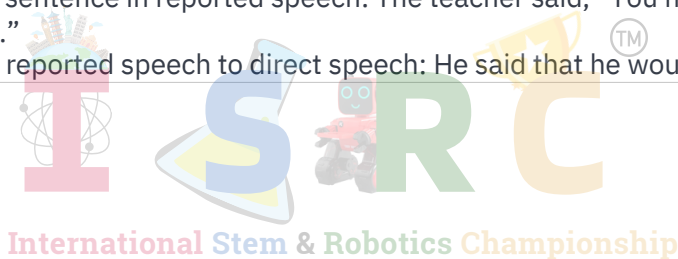
Reported speech (or indirect speech) is used to report what someone else has said without quoting them directly. It often requires changes in tense, pronouns, and time expressions.

Key Concepts:

- ☐ **Direct Speech:** Quoting the exact words spoken (e.g., He said, "I am tired.").
- ☐ **Reported Speech:** Paraphrasing what was said (e.g., He said that he was tired).

Questions:

1. Convert the direct speech to reported speech: "She said, „I am going to the market.”"
2. What changes occur when converting from direct to reported speech?
3. Rewrite this sentence in reported speech: The teacher said, "You must complete your homework."
4. Convert this reported speech to direct speech: He said that he would help us.



2. Mathematics

1.1 Number System

The number system is a way of classifying numbers based on their properties and the operations that can be performed on them.

Types of Numbers:

1. **Natural Numbers (N):** The set of positive integers (1, 2, 3, ...).
2. **Whole Numbers (W):** Natural numbers plus zero (0, 1, 2, 3, ...).
3. **Integers (Z):** Whole numbers and their negative counterparts (... , -3, -2, -1, 0, 1, 2, 3, ...).
4. **Rational Numbers (Q):** Numbers that can be expressed as a fraction (e.g., $\frac{1}{2}$, $-\frac{3}{4}$).
5. **Irrational Numbers:** Numbers that cannot be expressed as a fraction (e.g., $\sqrt{2}$, π).

Questions:

1. Identify three examples of irrational numbers.
2. What is the difference between whole numbers and natural numbers?
3. Convert the fraction $\frac{5}{10}$ into a decimal form.
4. Classify the number -7 as natural, whole, integer, rational, or irrational.

1.2 Polynomials

A polynomial is a mathematical expression consisting of variables, coefficients, and non-negative integer exponents.

Key Concepts:

- **Standard Form:** A polynomial written in descending order of degrees (e.g., $(3x^3 + 2x^2 - 5x + 4)$).
- **Types of Polynomials:**
 - **Monomial:** A polynomial with one term (e.g., $(4x^2)$).
 - **Binomial:** A polynomial with two terms (e.g., $(x^2 + 3x)$).
 - **Trinomial:** A polynomial with three terms (e.g., $(x^2 - x + 5)$).

Questions:

1. Identify whether the following is a polynomial: $(5x^3 - 4x + 7)$.
2. Write a polynomial in standard form with the terms $(3x^2, -2x, 6)$.
3. Classify the polynomial $(4x^3 + 3x - 7)$ as a monomial, binomial, or trinomial.
4. Evaluate the polynomial $(2x^2 + 3x - 5)$ for $(x = 2)$.

1.3 Coordinate Geometry

Coordinate geometry, or analytic geometry, involves the study of geometric figures using a coordinate system.

Key Concepts:

- ❑ **Cartesian Coordinate System:** A two-dimensional system defined by the x-axis (horizontal) and y-axis (vertical).
- ❑ **Points:** Represented as ordered pairs $((x, y))$.
- ❑ **Distance Formula:** The distance (d) between two points $((x_1, y_1))$ and $((x_2, y_2))$ is given by:
[
 $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
]

Questions:

1. What are the coordinates of the point that lies 3 units right and 2 units up from the origin?
2. Calculate the distance between the points (1, 2) and (4, 6).
3. How do you plot a point on the Cartesian plane?
4. Explain the significance of the x-axis and y-axis in coordinate geometry.

1.4 Linear Equation in Two Variables

A linear equation in two variables is an equation of the form $(Ax + By + C = 0)$, where A, B, and C are constants.

Key Concepts:

- ❑ **Graph of a Linear Equation:** A straight line on the Cartesian plane.
- ❑ **Solution of a Linear Equation:** Any ordered pair $((x, y))$ that satisfies the equation.

Questions:

1. Write a linear equation for the relationship: “y is twice x.”
2. Determine if the point (3, 6) is a solution for the equation $(2x + 3y = 24)$.
3. How do you graph a linear equation on the coordinate plane?
4. Explain the concept of slope in a linear equation.

1.5 Introduction to Euclid's Geometry

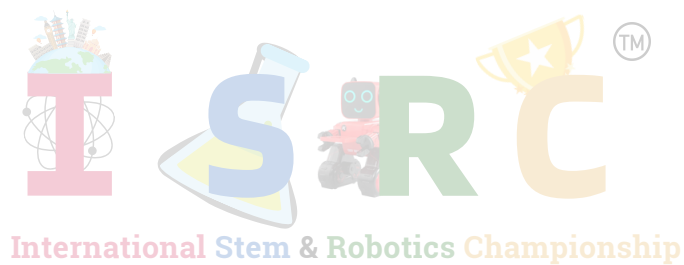
Euclidean geometry is the study of flat surfaces and shapes based on the postulates set forth by the ancient Greek mathematician Euclid.

Key Concepts:

- ❑ **Points, Lines, and Planes:** The basic undefined terms in geometry.
- ❑ **Postulates:** Statements accepted as true without proof (e.g., a straight line can be drawn between any two points).
- ❑ **Theorems:** Statements that can be proven based on postulates and previously established theorems.

Questions:

1. Define a point, line, and plane in geometry.
2. State and explain one of Euclid's postulates.
3. How do theorems differ from postulates?
4. Provide an example of a geometric figure and describe its properties using Euclidean concepts.



3.Science

1.1 Chemistry: Matter - Its Nature and Behaviour

Key Concepts:

- ❑ **Matter:** Anything that has mass and occupies space. Matter exists in different states: solid, liquid, and gas.
- ❑ **Properties of Matter:** Physical properties (e.g., color, melting point) and chemical properties (e.g., reactivity, flammability).

Examples:

- ❑ **Solid:** Ice, which has a fixed shape and volume.
- ❑ **Liquid:** Water, which has a definite volume but takes the shape of its container.
- ❑ **Gas:** Oxygen, which has neither a fixed shape nor volume.

Questions:

1. Define matter and give two examples of each state.
2. What is the difference between physical and chemical properties?
3. How does temperature affect the state of matter?
4. Describe the characteristics of solids, liquids, and gases.

1.2 Biology: Cell - Structure and Function

Key Concepts:

- ❑ **Cell:** The basic unit of life, which can exist as a single-celled organism or as part of a multicellular organism.
- ❑ **Cell Structure:** Major components include the cell membrane, cytoplasm, nucleus, and organelles (e.g., mitochondria, chloroplasts).

Examples:

- ❑ **Plant Cell:** Contains a cell wall, chloroplasts for photosynthesis, and a large central vacuole.
- ❑ **Animal Cell:** Lacks a cell wall, has smaller vacuoles, and contains lysosomes.

Questions:

1. What are the main parts of a cell and their functions?
2. How do plant cells differ from animal cells?
3. Describe the function of the nucleus in a cell.
4. Explain the importance of cell membranes.

1.3 Biology: Tissue

Key Concepts:

- **Tissue:** A group of cells that work together to perform a specific function. Tissues are classified into four main types: epithelial, connective, muscle, and nervous tissue.

Types of Tissues:

1. **Epithelial Tissue:** Covers body surfaces and lines cavities (e.g., skin).
2. **Connective Tissue:** Supports and binds other tissues (e.g., bone, blood).
3. **Muscle Tissue:** Responsible for movement (e.g., skeletal muscle, cardiac muscle).
4. **Nervous Tissue:** Transmits impulses for communication (e.g., brain, spinal cord).

Questions:

1. What are the four main types of tissues in the human body?
2. Describe the function of epithelial tissue.
3. How does connective tissue differ from muscle tissue?
4. Explain the role of nervous tissue in the body.

1.4 Physics: Force

Key Concepts:

- **Force:** A push or pull that can change the motion of an object. Forces are measured in newtons (N).
- **Types of Forces:**
 - **Contact Forces:** Require contact between objects (e.g., friction, tension).
 - **Non-contact Forces:** Act at a distance (e.g., gravitational force, magnetic force).

Examples:

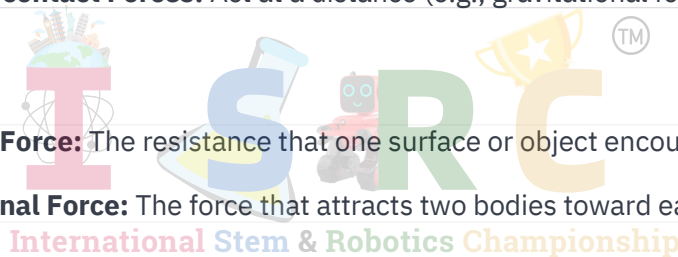
- **Frictional Force:** The resistance that one surface or object encounters when moving over another.
- **Gravitational Force:** The force that attracts two bodies toward each other.

Questions:

1. Define force and give two examples.
2. What is the difference between contact and non-contact forces?
3. How does friction affect the motion of an object?
4. Explain how gravitational force works.

1.5 Physics: Motion and Laws of Motion

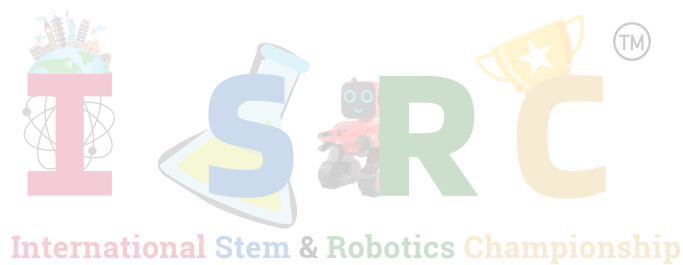
Key Concepts:



- **Motion:** The change in position of an object with respect to time. It can be described in terms of displacement, velocity, and acceleration.
- **Newton's Laws of Motion:**
 1. **First Law (Inertia):** An object at rest remains at rest, and an object in motion remains in motion unless acted upon by a net external force.
 2. **Second Law:** The acceleration of an object is directly proportional to the net force acting on it and inversely proportional to its mass ($F = ma$).
 3. **Third Law:** For every action, there is an equal and opposite reaction.

Questions:

1. What is the definition of motion?
2. State and explain Newton's First Law of Motion.
3. How does mass affect the acceleration of an object?
4. Provide an example of Newton's Third Law in everyday life.



4.Social Science

1.1 Geography: India - Size and Location

Key Concepts:

- ❑ **Size of India:** India is the seventh-largest country in the world, covering an area of approximately 3.287 million square kilometers.
- ❑ **Location:** India is located in South Asia, bordered by Pakistan, China, Nepal, Bhutan, Bangladesh, and Myanmar. It is surrounded by the Indian Ocean to the south.

Importance of Location:

- ❑ India's location influences its climate, agriculture, and trade routes.
- ❑ It is situated at the convergence of major trade routes, enhancing its historical significance in trade and cultural exchanges.

Questions:

1. What are the geographic coordinates of India?
2. Explain the significance of India's size and location in terms of biodiversity.
3. How does India's location influence its climate?
4. Discuss the impact of neighboring countries on India's foreign relations.

1.2 Geography: Physical Features of India

Key Concepts:

- ❑ **Landforms:** India has diverse physical features, including mountains, plains, plateaus, and deserts.
 - **Himalayas:** The highest mountain range, which includes peaks like Mount Everest.
 - **Indo-Gangetic Plain:** Fertile plains formed by the rivers Ganges, Yamuna, and Brahmaputra.
 - **Deccan Plateau:** A large plateau in southern India, known for its rich mineral resources.
 - **Thar Desert:** Located in the northwest, characterized by arid conditions.

Importance of Physical Features:

- ❑ The physical geography influences climate, vegetation, and human activities in India.

Questions:

1. Name the major mountain ranges in India.
2. What are the characteristics of the Indo-Gangetic Plain?
3. Describe the significance of the Deccan Plateau in terms of agriculture.
4. How do the physical features of India affect its climate?

1.3 Civics: What is Democracy? Why Democracy?

Key Concepts:

- ❑ **Democracy:** A system of government where the citizens exercise power directly or elect representatives to make decisions on their behalf.
- ❑ **Features of Democracy:**
 - **Political Equality:** Every citizen has the right to vote.
 - **Majority Rule:** Decisions are made based on the majority vote.
 - **Protection of Minority Rights:** Safeguards are in place to protect the rights of minority groups.

Importance of Democracy:

- ❑ Promotes participation, accountability, and transparency in governance.
- ❑ Ensures that the government reflects the will of the people.

Questions:

1. Define democracy in your own words.
2. Discuss the key features of a democratic government.
3. Why is it important to protect minority rights in a democracy?
4. Explain the advantages and disadvantages of democracy.

1.4 Economics: The Story of Village Palampur

Key Concepts:

- ❑ **Village Palampur:** A hypothetical village used to explain the economic activities and structures in rural India.
- ❑ **Economic Activities:**
 - **Agriculture:** The primary economic activity, with crops like wheat and rice being cultivated.
 - **Dairy Farming:** Important for additional income.
 - **Small-scale Industries:** Handicrafts and local businesses contribute to the economy.

Importance of Rural Economy:

- ❑ Rural economies are vital for the overall economic development of the country.
- ❑ Understanding village economics helps address issues like poverty and employment.

Questions:

1. Describe the main economic activities in Village Palampur.
2. How does agriculture influence the economy of rural areas?
3. Discuss the role of small-scale industries in village development.
4. Explain the challenges faced by farmers in rural areas.

1.5 History: The French Revolution

Key Concepts:

- ❑ **Causes of the French Revolution:** Social inequality, financial crisis, and the influence of Enlightenment ideas.
- ❑ **Key Events:**
 - **Storming of the Bastille:** Symbolic event marking the start of the revolution.
 - **Declaration of the Rights of Man and Citizen:** Established fundamental rights and freedoms.

Impact of the Revolution:

- ❑ Led to the rise of democracy and the end of feudalism in France.
- ❑ Inspired revolutions and movements for change worldwide.

Questions:

1. What were the main causes of the French Revolution?
2. Explain the significance of the Storming of the Bastille.
3. How did the French Revolution influence other countries?
4. Discuss the impact of Enlightenment ideas on the revolution.

1.6 History: Socialism in Europe

Key Concepts:

- ❑ **Socialism:** An economic and political system where the means of production are owned and regulated by the community or state.
- ❑ **Rise of Socialism:** Emerged in Europe as a response to the inequalities of capitalism and industrialization.

Key Developments:

- ❑ **Workers' Movements:** Labor unions and strikes aimed at improving working conditions and wages.
- ❑ **Political Parties:** Formation of socialist parties advocating for workers' rights and social reforms.

Questions:

1. What are the core principles of socialism?
2. How did industrialization contribute to the rise of socialism in Europe?
3. Discuss the impact of socialism on labor rights and movements.
4. Explain the differences between socialism and capitalism.

5. Mental Ability

1.1 Alphabet Test

Alphabet tests involve determining the position of letters in the English alphabet or finding relationships between different letters.

Examples:

1. **Position:** A = 1, B = 2, C = 3, ...
 - o **Question:** What is the position of the letter D?
☐
 - o **Answer:** 4
2. **Reverse Position:** Z = 1, Y = 2, X = 3, ...
 - o **Question:** What is the reverse position of the letter A?
☐
 - o **Answer:** 26

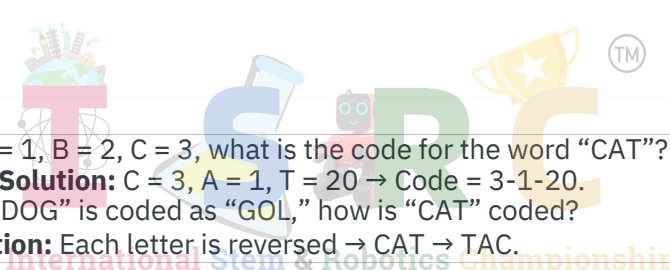
Questions:

1. What is the position of the letter F in the English alphabet?
2. If A = 1, what is the sum of the positions of the letters A, B, and C?
3. What is the reverse position of the letter M?
4. Find the letter that is 3 positions after J in the alphabet.

1.2 Coding-Decoding

Coding-decoding problems involve substituting letters or words based on a specific rule. These exercises test logical reasoning and pattern recognition.

Examples:

- 
1. **Code:** If A = 1, B = 2, C = 3, what is the code for the word "CAT"?
 - o **Solution:** C = 3, A = 1, T = 20 → Code = 3-1-20.
 2. **Code:** If "DOG" is coded as "GOL," how is "CAT" coded?
 - o **Solution:** Each letter is reversed → CAT → TAC.

Questions:

1. If A = 2, B = 4, C = 6, what is the code for "CAB"?
2. How would you code the word "FISH" if each letter is shifted by +1?
3. If "RAIN" is coded as "SBOJ," what is the code for "SUN"?
4. Decode the following: If M = 13, N = 14, O = 15, what does "MON" equal?

1.3 Series Completion

Series completion involves identifying the next item in a sequence based on a specific pattern.

Examples:

1. **Series:** 1, 3, 5, 7, ...
o **Next Number:** 9 (odd numbers).
2. **Series:** 2, 4, 8, 16, ...
o **Next Number:** 32 (each number is multiplied by 2).

Questions:

1. What is the next number in the series: 5, 10, 15, 20, ...? 2. Find the missing number in the series: 1, 4, __, 16, 25. 3. What comes next in the series: 3, 6, 12, 24, ...? 4. Identify the next term: A, B, D, F, ...?

1.4 Analogy & Similarity

Analogy problems involve comparing two different things to highlight their similarities. Similarity tests assess the likeness between pairs of items.

Examples:

1. **Analogy:** Hand is to Glove as Foot is to __?
o **Answer:** Sock.
2. **Analogy:** Bird is to Fly as Fish is to __?
o **Answer:** Swim.

Questions:

1. Teacher is to School as Doctor is to __? 2. Sun is to Day as Moon is to __? 3. Cat is to Kitten as Dog is to __? 4. What is the analogy for: Wheel is to Vehicle as Blade is to __?

1.5 Direction Sense Test

Direction sense problems involve understanding and interpreting directional information. These exercises test spatial awareness and reasoning.

Examples:

1. If you face north and turn 90° to the right, which direction are you facing?
o **Answer:** East.
2. You walk 10 meters south, then turn left and walk 5 meters. In which direction are you now?
o **Answer:** East.

Questions:

1. If you are facing west and turn 180°, which direction are you facing now?
2. You walk 5 km north, then 3 km east. How far are you from your starting point?

3. If you go south from your house, turn right, and then go east, which direction are you facing now?
4. A person walks 2 km north and then 2 km east. What is the shortest distance back to the starting point?

1.6 Logical Venn Diagrams

Venn diagrams are used to illustrate the relationships between different sets. They help visualize the commonalities and differences.

Examples:

1. **Venn Diagram:** Show the relationship between “Mammals,” “Animals,” and “Pets.”
2. **Question:** If Set A = {1, 2, 3} and Set B = {2, 3, 4}, what is the intersection of Set A and Set B?

Questions:

1. Draw a Venn diagram to represent the sets of “Fruits” and “Vegetables.”
2. If Set X = {A, B, C} and Set Y = {B, C, D}, what is the union of Set X and Set Y?
3. What is the difference between the two sets A and B if A = {1, 2, 3, 4} and B = {2, 4, 6}?
4. Explain how Venn diagrams can help in understanding set theory.

1.7 Dice

Dice problems involve understanding the possible outcomes when rolling one or more dice. These exercises help in probability and combinatorics.

Examples:

1. If a die is rolled, what is the probability of rolling a number greater than 4?
o **Answer:** 1/3 (only 5 and 6).
2. If two dice are rolled, what is the probability of getting a sum of 7?
o **Answer:** 6/36 or 1/6.

International Stem & Robotics Championship

Questions:

1. How many faces does a standard die have?
2. What is the probability of rolling an even number on a single die?
3. If two dice are rolled, what is the total number of possible outcomes?
4. Explain how to calculate the probability of a specific outcome when rolling two dice.

1.8 Arithmetical Problems

Arithmetical problems involve mathematical calculations and the application of basic arithmetic operations.

Examples:

1. **Problem:** If you buy 3 apples for \$2 each, how much do you spend in total?
o **Solution:** $3 \times 2 = \$6$.
2. **Problem:** What is 25% of 200?
o **Solution:** $0.25 \times 200 = 50$.

Questions:

1. If you have 15 candies and you give away 5, how many do you have left?
2. Calculate the sum of 234 and 567.
3. If a book costs \$15 and you buy 4, what is the total cost?
4. Solve the equation: $5x + 3 = 18$.

